Real Numbers: Mini-Lessons

NC.8.NS.1–2 · NC.8.EE.2

Name _____ Date _____ Period _____

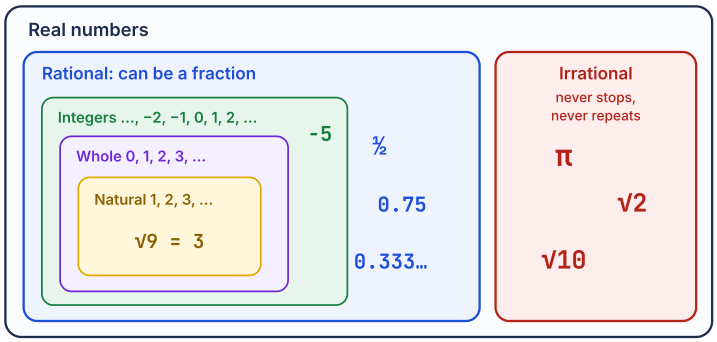
1.1 Rational or irrational? NC.8.NS.1

A number is **rational** if it can be written as a **fraction** of two integers. Its decimal **stops** or **repeats**. A number is **irrational** if its decimal **never stops and never repeats**, like π or √2.

Terminating decimal
a decimal that stops: 0.75

Repeating decimal
a decimal with digits that repeat forever: 0.333...

Perfect square
a whole number times itself: 1, 4, 9, 16, 25, 36, 49, ...



Square roots: √(perfect square) is rational (√49 = 7). √(not a perfect square) is irrational (√10).
Say it this way: "___ is rational because it can be written as a fraction."

Watch out: A long decimal is not always irrational. If it **repeats**, it is rational.

Walk through: Is √49 rational or irrational?

1. Is 49 a perfect square?

2. So what is √49?

3. Can 7 be written as a fraction?

Check your understanding

1 Score 3 NC.8.NS.1

Is 0.333… rational or irrational?

Ⓐ Rational Ⓑ Irrational

2 Score 3 NC.8.NS.1

Is √10 rational or irrational?

Ⓐ Irrational Ⓑ Rational

3 Score 3 NC.8.NS.1

Which number is **irrational**?

(A) π (B) −5

(C) 0.75 (D) √16

1.2 Repeating decimals to fractions (the 9's rule) NC.8.NS.1

To change a repeating decimal into a fraction: the **repeating digits go on top**. The **bottom is all 9s**, one 9 for each repeating digit. Then **simplify**.

Bar notation

a bar over the digits that repeat: $0.\overline{3} = 0.333...$

Simplify

divide the top and bottom by the same number

$0.272727\ldots$ $0.\overline{27}$

$$\frac{27}{99} \div 9 = \frac{3}{11}$$

2 repeating digits → two 9s

Decimal	Repeating digits	Fraction	Simplified
$0.\overline{3}$	1 digit	3/9	1/3
$0.\overline{27}$	2 digits	27/99	3/11
$0.\overline{125}$	3 digits	125/999	125/999

Watch out: Count the 9s carefully: one 9 for **each** repeating digit. And always simplify.

Walk through: Write 0.363636… as a fraction

1 Which digits repeat?

2 What is the bottom of the fraction?

3 Simplify 36/99. What number divides both?

Check your understanding

1 Score 3 NC.8.NS.1

Write $0.\overline{5}$ as a fraction.

(A) 5/10 (B) 5/9

(C) 1/5 (D) 5/99

2 Score 3 NC.8.NS.1

Write $0.\overline{12}$ as a fraction in simplest form.

(A) 12/9 (B) 4/33

(C) 3/25 (D) 12/99

1.3 Square roots and cube roots NC.8.EE.2

A **square root** undoes squaring: $\sqrt{25} = 5$ because $5 \times 5 = 25$. A **cube root** undoes cubing: $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$.

Squared (^{2})

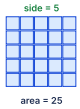
a number times itself: $6^2 = 6 \times 6 = 36$

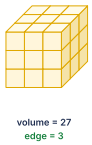
Cubed (^{3})

a number times itself 3 times: $2^3 = 2 \times 2 \times 2 = 8$

$\sqrt{}$ and $\sqrt[3]{}$

square root and cube root: they go backward





n	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	125	216	343	512	729	1000

Watch out: $\sqrt{}$ does **not** mean "divide by 2", and $\sqrt[3]{}$ does **not** mean "divide by 3". $\sqrt{16} = 4$, not 8.

Walk through: Find $\sqrt[3]{125}$

1 What does $\sqrt[3]{125}$ ask?

2 Try 5. What is $5 \times 5 \times 5$?

3 So $\sqrt[3]{125} = ?$

Check your understanding

1 Score 3 NC.8.EE.2

Find $\sqrt{81}$.

Answer: _____

2 Score 3 NC.8.EE.2

Find $\sqrt[3]{8}$.

Answer: _____

3 Score 3 NC.8.NS.1

Which number is a **perfect square**?

(A) 50 (B) 18

(C) 27 (D) 36

1.4 Estimate roots and order numbers NC.8.NS.2

To estimate a square root that is not a perfect square, find the **perfect squares just below and just above**. The root is **between** their roots. To compare numbers, write them all as **decimals** first.

Estimate

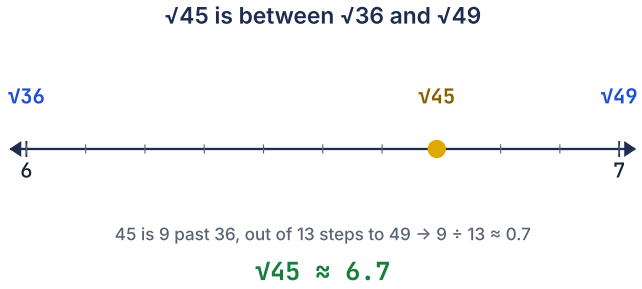
a close guess you can explain

Between

√45 is between 6 and 7 because 36 < 45 < 49

Least to greatest

smallest first, biggest last



4-step estimate of √45: 36 < 45 < 49 → (45 − 36) ÷ (49 − 36) = 9 ÷ 13 ≈ 0.7 → √45 ≈ 6.7

Watch out: Line up the decimals before you compare. 3.5 is bigger than 3.375 even though "375" looks long.

Walk through: `Estimate √20 to the nearest tenth`

1. Which perfect squares are just below and just above 20?

_____

2. So √20 is between...

_____

3. Find the tenths: (20 − 16) ÷ (25 − 16) = ?

_____

Check your understanding

1 Score 3 NC.8.NS.2

$\sqrt{30}$ is between which two whole numbers?

(A) 5 and 6
(B) 15 and 16
(C) 6 and 7
(D) 25 and 36

2 Score 3 NC.8.NS.2

Which is greater: $\sqrt{50}$ or 7.5?

(A) 7.5
(B) √50
(C) They are equal

3 Score 3 NC.8.NS.2

Order from least to greatest: $\sqrt{5}, 2.1, 7/3$

(A) √5, 2.1, 7/3
(B) 2.1, 7/3, √5
(C) 7/3, √5, 2.1
(D) 2.1, √5, 7/3

SKILL BUILDERS · UNIT 1 · SCORE 3

Real Numbers: Practice Set

NC.8.NS.1–2 · NC.8.EE.2

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.NS.1

Is $0.1212\ldots$ rational or irrational?

(A) Rational (B) Irrational

2 Score 3 NC.8.NS.1

Is $\sqrt{144}$ rational or irrational?

(A) Rational (B) Irrational

3 Score 3 NC.8.NS.1

Is $\sqrt{8}$ rational or irrational?

(A) Irrational (B) Rational

4 Score 3 NC.8.NS.1

Write $0.\overline{7}$ as a fraction.

(A) 7/9 (B) 7/10

(C) 1/7 (D) 7/99

5 Score 3 NC.8.NS.1

Write $0.\overline{45}$ as a fraction in simplest form.

(A) 45/100 (B) 5/11

(C) 45/99 (D) 9/20

6 Score 3 NC.8.EE.2

Find $\sqrt{121}$.

Answer: _____

7 Score 3 NC.8.EE.2

Find $\sqrt[3]{64}$.

Answer: _____

8 Score 3 NC.8.NS.2

$\sqrt{40}$ is between which two whole numbers?

(A) 20 and 21 (B) 6 and 7

(C) 5 and 6 (D) 36 and 49

9 Score 3 NC.8.NS.2

Estimate $\sqrt{50}$ to the nearest tenth.

Answer: _____

10 Score 3 NC.8.NS.2

Order from least to greatest:

$\pi,\ \sqrt{10},\ 10/3,\ 3.5$

(A) √10, π, 10/3, 3.5

(B) π, √10, 10/3, 3.5

(C) 3.5, 10/3, √10, π

(D) π, 10/3, √10, 3.5

11 Score 3 NC.8.EE.2

Find the mistake. Sam wrote: **√36 = 18**. What went wrong?

(A) Sam divided 36 by 2. √36 = 6 because 6 × 6 = 36.

(B) Sam is right: 18 + 18 = 36.

(C) √36 = 9 because 4 × 9 = 36.

(D) √36 = 72 because 36 × 2 = 72.

SKILL BUILDERS · UNIT 2 · SCORE 3

Exponents & Scientific Notation: Mini-Lessons

NC.8.EE.1, 3–4

Name _____ Date _____ Period _____

2.1 What an exponent means NC.8.EE.1

An **exponent** is a shortcut for multiplying the same number over and over. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Base
the number being multiplied (in 2^5, the 2)

Exponent
how many times the base is multiplied (the 5)

Expanded form
written out the long way: 2 × 2 × 2 × 2 × 2

base exponent

$$2^5$$

$$2 \times 2 \times 2 \times 2 \times 2 = 32$$

NOT 2 × 5 = 10

Parentheses matter: $(-3)^2 = (-3)(-3) = 9$, but $-3^2 = -(3 \times 3) = -9$.

Watch out: 2^5 is **not** 2 × 5. Write it out to check.

Walk through: Find $(-2)^3$

1. Write $(-2)^3$ in expanded form.

2. Multiply the first two: $(-2)(-2)$ = ?

3. Now $4 \cdot (-2)$ = ?

Check your understanding

1 Score 3 NC.8.EE.1

Find the value of 3^4.

Answer: _____

2 Score 3 NC.8.EE.1

Which is the same as 5^3?

(A) 5 + 5 + 5
(B) 5 × 3
(C) 3 × 3 × 3 × 3 × 3
(D) 5 × 5 × 5

3 Score 3 NC.8.EE.1

Find the value of $(-4)^2$.

Answer: _____

2.2 Multiply and divide powers NC.8.EE.1

Same base: when you **multiply**, **add** the exponents. When you **divide**, **subtract** the exponents. $x^a \cdot x^b = x^{a+b}$ and $x^a \div x^b = x^{a-b}$.

Same base
the numbers or letters being raised to powers match: 2^3 and 2^4

Coefficient
the number in front: in $3x^2$, the 3

$$x^3 \cdot x^4$$

$$(x \cdot x \cdot x) \cdot (x \cdot x \cdot x \cdot x)$$

7 x's multiplied

$$x^3 \cdot x^4 = x^{3+4} = x^7$$

Multiply same base → ADD the exponents

$2^3 \times 2^4 = 2^{3+4} = 2^7$

$x^5 \div x^2 = x^{5-2} = x^3$

$(3x^2)(2x^4) = 6x^6$ (multiply the numbers, add the exponents)

Watch out: Do **not** multiply the bases. $2^3 \times 2^4$ is 2^7, not 4^7.

Walk through: Simplify $(4x^3)(2x^5)$

1. First, the numbers in front. 4 × 2 = ?

_____

2. Next, the x's: $x^3 \cdot x^5$. What do we do with the exponents?

_____

3. Put it together.

_____

Check your understanding

1 Score 3 NC.8.EE.1

Simplify: $5^2 \times 5^5$

(A) 5^{10} (B) 5^7

(C) 5^3 (D) 25^7

2 Score 3 NC.8.EE.1

Simplify: $a^7 \div a^3$

(A) a^4 (B) $a^{7/3}$

(C) a^{21} (D) a^{10}

2.3 Power of a power, zero and negative exponents NC.8.EE.1

Power of a power: multiply the exponents, $(x^3)^2 = x^6$. **Zero exponent:** any nonzero number to the 0 power is **1**. **Negative exponent:** flip it to a fraction, $2^{-3} = 1/2^3 = 1/8$.

Reciprocal
the flip of a number: the reciprocal of 5 is 1/5

Power of a power
a power raised to another power, like $(x^3)^2$

Rule	Example
$(x^a)^b = x^{a \cdot b}$	$(3^2)^3 = 3^6$
$x^0 = 1$	$9^0 = 1$
$x^{-a} = 1/x^a$	$10^{-2} = 1/100$

$2^3 = 8$
$2^2 = 4$ ÷ 2
$2^1 = 2$ ÷ 2
$2^0 = 1$ ÷ 2
$2^{-1} = 1/2$ ÷ 2
$2^{-2} = 1/4$ ÷ 2

Watch out: A negative exponent does **not** make the number negative. $2^{-3} = 1/8$, not -8.

Walk through: Write 5^{-2} as a fraction

1 The exponent is negative. What does that tell us?

2 What is 5^2?

Check your understanding

1 Score 3 NC.8.EE.1

Simplify: $(x^4)^2$

(A) x^2 (B) x^8

(C) x^6 (D) x^{16}

2 Score 3 NC.8.EE.1

Find the value of 9^0.

Answer: _____

3 Score 3 NC.8.EE.1

Find the value of 2^{-3}. (Type a fraction like 1/4.)

Answer: _____

2.4 Scientific notation NC.8.EE.3

Scientific notation writes a number as $a \times 10^n$, where **a is at least 1 and less than 10**. Big numbers get a **positive** exponent. Small numbers (less than 1) get a **negative** exponent.

Front number (a)

must be 1 or more, and less than 10: 4.5 ✓, 45 ×, 0.45 ×

Power of 10

the exponent counts how many places the decimal point moves



$0.00032 = 3.2 \times 10^{-4}$

front number 4.5 is between 1 and 10 ✓

Standard form	Hops	Scientific notation
45,000,000	7 left	4.5×10^7
0.00032	4 right	3.2×10^{-4}
6,020,000	6 left	6.02×10^6

Watch out: 45×10^3 is not finished: 45 is not less than 10. Fix it to 4.5×10^4.

Walk through: `Write 0.0061 in scientific notation`

1. Where should the decimal point go so the front number is between 1 and 10?

2. How many places did the point hop?

3. Positive or negative exponent?

Check your understanding

1 Score 3 NC.8.EE.3

Write **32,000** in scientific notation.

(A) 3.2×10^{-4} (B) 3.2×10^{3}

(C) 3.2×10^{4} (D) 32×10^{3}

2 Score 3 NC.8.EE.3

Write 5×10^{3} in standard form.

Answer: _____

2.5 Multiply and divide in scientific notation NC.8.EE.4

Multiply: multiply the front numbers and **add** the exponents. **Divide:** divide the front numbers and **subtract** the exponents. Then check that the front number is between 1 and 10.

Adjust

if the front number is 10 or more, move the point and fix the exponent: $15 \times 10^{7} = 1.5 \times 10^{8}$

$(8 \times 10^{7}) \div (4 \times 10^{3}) = (8 \div 4) \times 10^{7-3} = 2 \times 10^{4}$

$(3 \times 10^{4})(2 \times 10^{5})$

$= (3 \times 2) \times 10^{4+5}$

$= 6 \times 10^{9}$

front numbers: multiply · tens: add exponents

Watch out: After multiplying, check the front number. 15×10^{7} must be fixed to 1.5×10^{8}.

Walk through: `Multiply (5 × 10³)(3 × 10⁴)`

1. Multiply the front numbers.

2. Now the powers of 10: $10^{3} \cdot 10^{4} = ?$

3. 15 is not less than 10. Fix it.

Check your understanding

1 Score 3 NC.8.EE.4

Multiply: $(2 \times 10^3)(4 \times 10^2)$

(A) 8×10^6

(B) 8×10^1

(C) 8×10^5

(D) 6×10^5

2 Score 3 NC.8.EE.4

Divide: $(9 \times 10^8) \div (3 \times 10^2)$

(A) 3×10^6

(B) 3×10^4

(C) 6×10^6

(D) 3×10^{10}

SKILL BUILDERS · UNIT 2 · SCORE 3

Exponents & Scientific Notation: Practice Set

NC.8.EE.1, 3–4

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.EE.1

Simplify: $4^3 \times 4^2$

(A) 16^5 (B) 4^5

(C) 4^6 (D) 4^1

2 Score 3 NC.8.EE.1

Simplify: $x^7 \div x^4$

(A) $x^{7/4}$ (B) x^{11}

(C) x^{28} (D) x^3

3 Score 3 NC.8.EE.1

Simplify: $(2^3)^2$

(A) 2^6 (B) 4^3

(C) 2^9 (D) 2^5

4 Score 3 NC.8.EE.1

Find the value of 5^0.

Answer: _____

5 Score 3 NC.8.EE.1

Find the value of 3^{-2}. (Type a fraction like 1/4.)

Answer: _____

6 Score 3 NC.8.EE.1

Find the value of $(-3)^2$.

Answer: _____

7 Score 3 NC.8.EE.3

Write 0.00056 in scientific notation.

(A) 5.6×10^{-3} (B) 5.6×10^{-4}

(C) 56×10^{-5} (D) 5.6×10^{4}

8 Score 3 NC.8.EE.3

Write 7.2×10^6 in standard form.

Answer: _____

9 Score 3 NC.8.EE.4

Multiply: $(5 \times 10^3)(3 \times 10^4)$

(A) 1.5×10^8 (B) 1.5×10^7

(C) 15×10^7 (D) 8×10^7

10 Score 3 NC.8.EE.4

Divide: $(8 \times 10^7) \div (4 \times 10^3)$

(A) 2×10^{10} (B) 2×10^4

(C) 4×10^4 (D) 32×10^4

11 Score 3 NC.8.EE.1

Find the mistake. Lee wrote: $x^3 \cdot x^4 = x^{12}$. What went wrong?

(A) Lee multiplied the exponents. Multiplying powers adds them: x^7.

(B) Lee is right.

(C) It should be $2x^7$.

(D) It should be x^1 because $4 - 3 = 1$.

Equations & Inequalities: Mini-Lessons

NC.8.EE.7

Name _____ Date _____ Period _____

3.1 One- and two-step equations NC.8.EE.7

To **solve** an equation, get **x alone**. Undo each move with its **opposite**, and do it to **both sides** so the equation stays balanced.

Variable
a letter, like x, that stands for a number

Inverse (opposite) operation
the move that undoes another: + and − undo each other, × and ÷ undo each other

Solution
the number that makes the equation true

To undo...	+ 5	− 4	× 3 (3x)	÷ 2
do this	− 5	+ 4	÷ 3	× 2

Order: undo + or − first, then × or ÷.

Watch out: Do the same move to **both sides**. A move on one side only breaks the balance.

x = 4

Check: 2(4) + 3 = 11 ✓



Walk through: `Solve 3x - 4 = 11`

1. Which move comes first?

_____

2. x is multiplied by 3. What undoes that?

_____

3. Check: put 5 in for x. What is 3(5) − 4?

_____

Check your understanding

1 Score 3 NC.8.EE.7

Solve: $x + 7 = 15$

Answer: _____

2 Score 3 NC.8.EE.7

What is the **first** move to solve $4x + 5 = 21$?

(A) Add 5 to both sides

(B) Divide both sides by 4

(C) Subtract 5 from both sides

(D) Subtract 4 from both sides

3 Score 3 NC.8.EE.7

Solve: $2x + 3 = 11$

Answer: _____

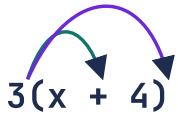
3.2 Distribute and combine like terms NC.8.EE.7

The number outside the parentheses **multiplies every term inside**. **Like terms** (x-terms with x-terms, numbers with numbers) can be added together.

Distribute
multiply the outside number by each term inside the ( )

Like terms
terms with the same variable part: 3x and 5x, or 4 and 9

Coefficient
the number in front of a variable (in 3x, the 3)



$3 \cdot x + 3 \cdot 4$

$3x + 12$

$4(x - 2) = 4x - 8$

$3x + 5 + 2x = 5x + 5$

Watch out: 3(x + 4) is **not** 3x + 4. The 3 multiplies the 4 too: 3x + 12.

Walk through: Solve $2(x + 5) = 16$

1 There are parentheses. What is the first move?

_____

2 Undo the + 10.

3 Get x alone.

Check your understanding

1 Score 3 NC.8.EE.7

Which is equal to $3(x + 4)$?

(A) 3x + 4 (B) x + 12

(C) 3x + 12 (D) 3x + 7

2 Score 3 NC.8.EE.7

Combine like terms: $5x + 2 + 3x$

(A) $8x^2$ (B) 15x + 2

(C) 10x (D) 8x + 2

3 Score 3 NC.8.EE.7

Solve: $4(x - 2) = 20$

Answer: _____

3.3 Variables on both sides NC.8.EE.7

When x is on **both sides**, move the x-terms to one side first (subtract the smaller x-term from both sides). Then solve the two-step equation that is left.

Like terms

5x and 3x are like terms, so 5x − 3x = 2x

Both sides

the left of the = sign and the right of the = sign

The 5-step plan:

1. Distribute (if there are parentheses).
2. Combine like terms on each side.
3. Move the x-terms to one side.
4. Move the numbers to the other side.
5. Divide to get x alone.

Watch out: Subtract the x-term from **both** sides, not just the side it is on.

1 $5x + 2 = 3x + 10$ → Subtract 3x
$-3x \quad -3x$

2 $2x + 2 = 10$ → Subtract 2
$-2 \quad -2$

3 $\frac{2x}{2} = \frac{8}{2}$ → Divide by 2

4 $x = 4$

Walk through: Solve $7x + 3 = 4x + 15$

1 x is on both sides. What move puts the x's together?

2 Undo the + 3.

3 Get x alone.

Check your understanding

1 Score 3 NC.8.EE.7

What is a good **first** move for $6x - 1 = 2x + 11$?

(A) Subtract 2x from both sides

(B) Divide both sides by 6

(C) Add 2x to both sides

(D) Subtract 6x from the right side only

2 Score 3 NC.8.EE.7

Solve: $6x - 1 = 2x + 11$

Answer: _____

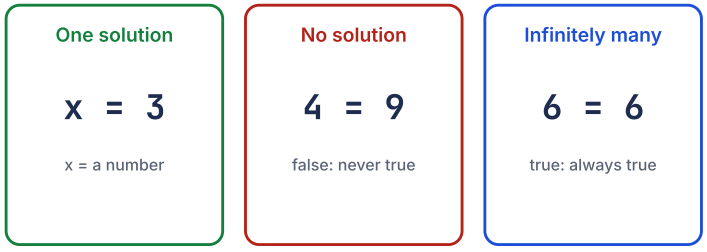
3.4 One, none, or infinitely many solutions NC.8.EE.7a

Solve as usual. If the x's cancel, look at what is left. **False** (like 4 = 9) means **no solution**. **True** (like 6 = 6) means **infinitely many solutions**. If you get x = a number, there is **one solution**.

No solution
no number makes the equation true

Infinitely many solutions
every number makes the equation true

One solution
exactly one number works, like x = 3



$2x + 4 = 2x + 9 \rightarrow 4 = 9$ (false) → no solution

$3(x + 2) = 3x + 6 \rightarrow 6 = 6$ (true) → infinitely many

Watch out: If the x's cancel, you are not stuck. The last line tells you the answer.

Walk through: How many solutions? 3(x + 2) = 3x + 6

1. First move?

2. Subtract 3x from both sides. What is left?

3. 6 = 6 is true. How many solutions?

Check your understanding

1 Score 3 NC.8.EE.7a

How many solutions does **2x + 4 = 2x + 9** have?

(A) No solution
(B) One solution
(C) Infinitely many solutions

2 Score 3 NC.8.EE.7a

A last line says **5 = 5**. How many solutions?

(A) Infinitely many solutions
(B) One solution
(C) No solution

3 Score 3 NC.8.EE.7a

How many solutions does **4x + 1 = 2x + 7** have?

(A) One solution
(B) No solution
(C) Infinitely many solutions

3.5 Inequalities: solve and graph NC.8.EE.7

Solve an inequality like an equation, with **one special rule**: when you **multiply or divide by a negative number, flip the sign**. Then graph it on a number line.

< and >

less than, greater than: open circle (the number is not included)

≤ and ≥

at most, at least: closed circle (the number is included)

Flip

turn the sign around: < becomes >, ≤ becomes ≥



x > 2: open circle at 2, shade right

Watch out: Only flip when you **multiply or divide by a negative**. Adding or subtracting never flips.

$-2x < 6$

$\div(-2) \quad \div(-2)$

$x > -3$



Walk through: Solve 2x + 1 ≥ 7 and graph it

1. Undo the + 1.

2. Undo the × 2. Do we flip?

3. Open or closed circle at 3?

4. Which way do we shade?

Check your understanding

1 Score 3 NC.8.EE.7

Solve: $x + 4 > 9$

(A) x ≥ 5
(B) x > 5
(C) x < 5
(D) x > 13

2 Score 3 NC.8.EE.7

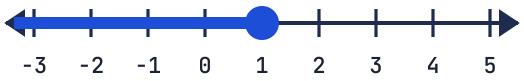
Solve: $-2x < 8$

(A) x > −4
(B) x < −4
(C) x < −16
(D) x > 4

3 Score 3 NC.8.EE.7

Which graph shows $x \leq 1$?

A



B



C



SKILL BUILDERS · UNIT 3 · SCORE 3

Equations & Inequalities: Practice Set

NC.8.EE.7

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.EE.7

Solve: $x - 7 = 10$

Answer: _____

2 Score 3 NC.8.EE.7

Solve: $3x = 18$

Answer: _____

3 Score 3 NC.8.EE.7

Solve: $2x + 5 = 19$

Answer: _____

4 Score 3 NC.8.EE.7

Solve: $4x - 3 = 25$

Answer: _____

5 Score 3 NC.8.EE.7

Solve: $4(x + 1) = 24$

Answer: _____

6 Score 3 NC.8.EE.7

Solve: $5x - 2 = 3x + 8$

Answer: _____

7 Score 3 NC.8.EE.7

Solve: $3(x - 2) + x = 10$

Answer: _____

8 Score 3 NC.8.EE.7

How many solutions does $2x + 6 = 2x + 1$ have?

(A) No solution

(B) One solution

(C) Infinitely many solutions

9 Score 3 NC.8.EE.7

Solve: $-3x \leq 12$

(A) $x \geq -4$

(B) $x \geq 4$

(C) $x \leq -4$

(D) $x \leq -36$

10 Score 3 NC.8.EE.7

Which graph shows $x + 4 > 9$?

(A)


(B)


(C)


(D)


11 Score 3 NC.8.EE.7

Find the mistake. Jordan solved $2(x + 3) = 14$:

Line 1: $2x + 3 = 14$

Line 2: $2x = 11$

Line 3: $x = 5.5$

Where is the mistake?

(A) Line 1: the 2 was not multiplied by 3. It should be 2x + 6 = 14.

(B) Line 2: 14 − 3 is not 11.

(C) There is no mistake.

(D) Line 3: 11 ÷ 2 is not 5.5.

12 Score 3 NC.8.EE.7

Five more than twice a number is 17. The equation is **2x + 5 = 17**. What is the number?

Answer: _____

SKILL BUILDERS · UNIT 4 · SCORE 3

Angle Relationships: Mini-Lessons

NC.8.G.5

Name _____ Date _____ Period _____

4.1 Angle pairs NC.8.G.5

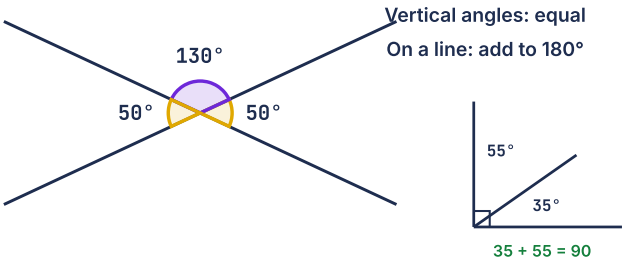
Angles come in pairs that follow rules. **Complementary** angles add to **90°**. **Supplementary** angles add to **180°**. **Vertical** angles (across from each other where two lines cross) are **equal**.

Complementary
two angles that add to 90° (a corner)

Supplementary
two angles that add to 180° (a straight line)

Vertical angles
across from each other at a crossing: equal

Adjacent
next to each other, sharing a side



Watch out: Mixing up 90 and 180?
Complementary = **C**orner = 90°.
Supplementary = **S**traight = 180°.

Walk through: `Two angles are supplementary. One is 110°. Find the other.`

1 Supplementary angles add to...

_____

2 How do we find the missing angle?

_____

3 The other angle is...

_____

Check your understanding

1 Score 3 NC.8.G.5

Find the **complement** of 62°.

Answer: _____ °

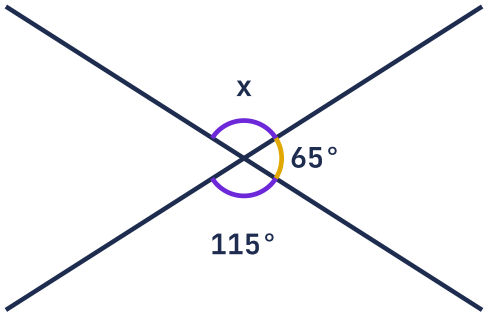
2 Score 3 NC.8.G.5

Find the **supplement** of 145°.

Answer: _____ °

3 Score 3 NC.8.G.5

Two lines cross. Find **x**.



Answer: _____ °

4.2 Parallel lines cut by a transversal NC.8.G.5

When a transversal cuts two **parallel** lines, every angle is either **equal** to a given angle or **adds with it to 180°**. Equal pairs: **corresponding, alternate interior, alternate exterior, vertical**. Same-side interior angles add to **180°**.

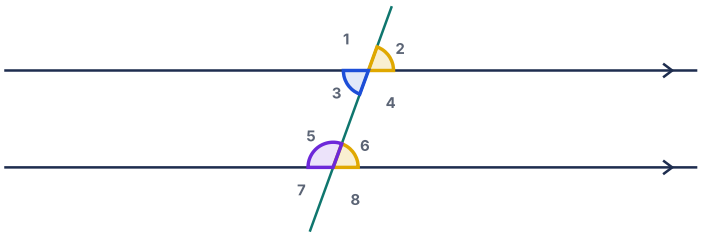
Transversal
a line that crosses two other lines

Corresponding
same corner at each crossing: equal

Alternate interior
between the lines, opposite sides: equal

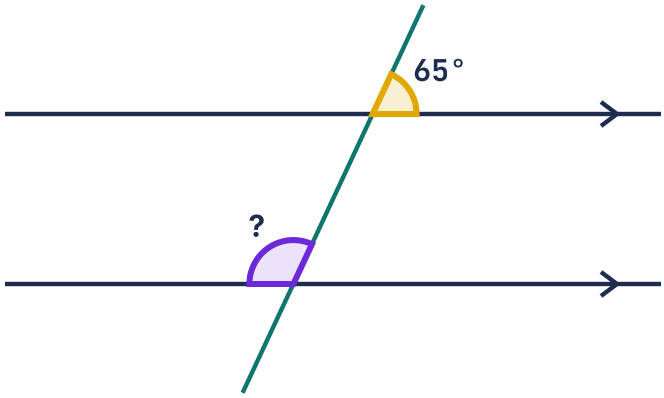
Same-side interior
between the lines, same side: add to 180°

Watch out: Look at the picture: a small-looking angle can only equal another small-looking angle.



small = 70°, big = 110°

Walk through: `Lines are parallel. ∠2 = 65°. Find ∠5.`



1 Is ∠5 small-looking (acute) or big-looking (obtuse)?

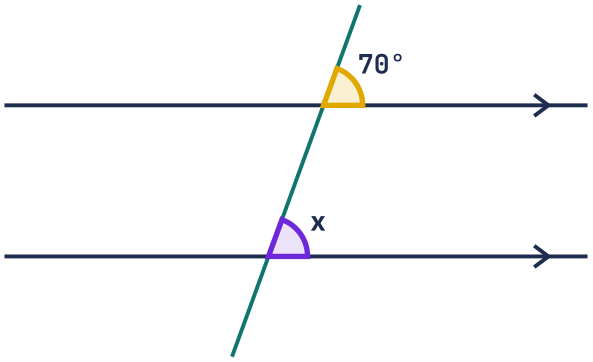
2 So ∠5 and ∠2 are...

3 Find ∠5.

Check your understanding

1 Score 3 NC.8.G.5

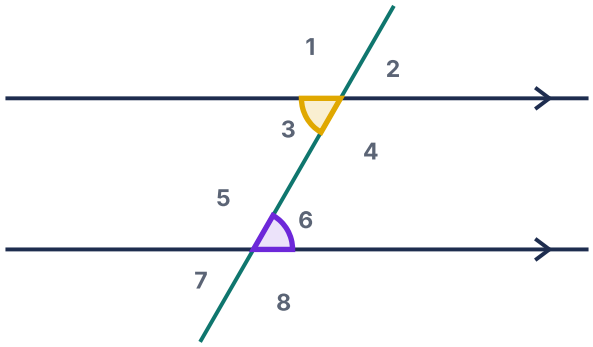
The lines are parallel. Find **x**.



Answer: _____ °

2 Score 3 NC.8.G.5

What kind of pair are ∠3 and ∠6?



Ⓐ Corresponding angles

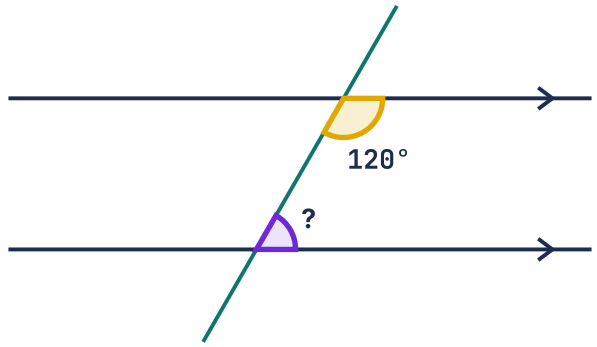
Ⓑ Alternate interior angles

Ⓒ Vertical angles

Ⓓ Same-side interior angles

3 Score 3 NC.8.G.5

The lines are parallel. ∠4 = 120°. Find ∠6 (a same-side interior angle).



Answer: _____ °

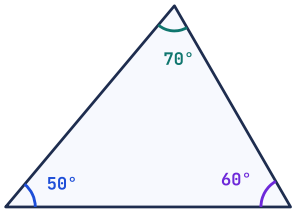
4.3 Angles in a triangle NC.8.G.5

The three inside angles of **any** triangle add to **180°**. To find a missing angle: add the two you know, then subtract from 180.

Interior angle
an angle inside the triangle

Right triangle
a triangle with one 90° angle (a square corner)

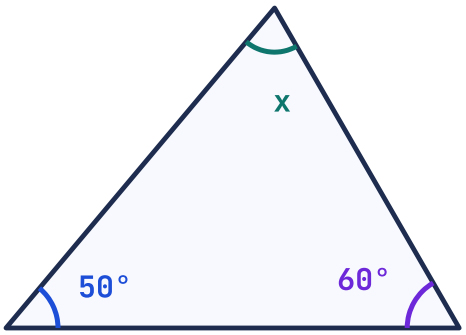
Watch out: Use 180°, not 360°. 360° is a full circle (or a four-sided shape).



50 + 70 + 60 = 180



Walk through: `A triangle has angles 50° and 60°. Find the third angle.`



1 The three angles of a triangle add to...

_____

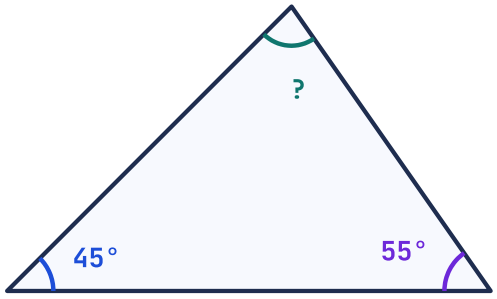
2 Add the angles you know.

3 Find x.

Check your understanding

1 Score 3 NC.8.G.5

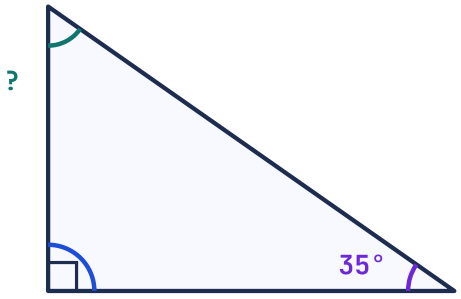
A triangle has angles 45° and 55°. Find the third angle.



Answer: _____ °

2 Score 3 NC.8.G.5

A right triangle has one angle of 35°. Find the other small angle.



Answer: _____ °

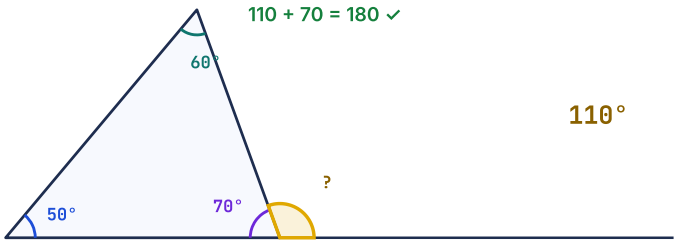
4.4 The exterior angle of a triangle NC.8.G.5

An **exterior angle** is outside the triangle, made by stretching one side. It **equals the two far inside angles added together**.

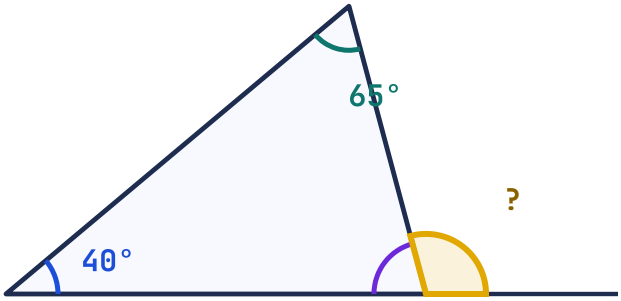
Exterior angle
the angle outside the triangle, next to one inside angle

Remote (far) interior angles
the two inside angles that do not touch the exterior angle

Watch out: Use the two **far** angles, not the angle right next to the exterior angle.



Walk through: `The far inside angles are 40° and 65°. Find the exterior angle.`



1 Which angles do we use?

_____

2 Add them.

_____

Check your understanding

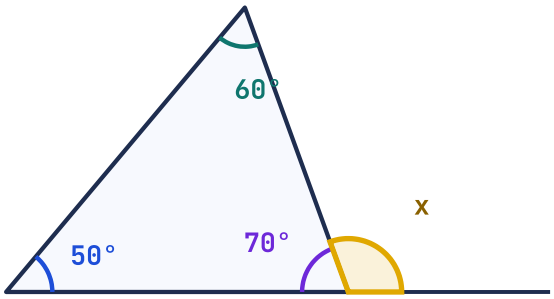
1 Score 3 NC.8.G.5

The far inside angles are 40° and 65°. Find the exterior angle.

Answer: _____ °

2 Score 3 NC.8.G.5

Find the exterior angle **x**.



Answer: _____ °

SKILL BUILDERS · UNIT 4 · SCORE 3

Angle Relationships: Practice Set

NC.8.G.5

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.5

Find the **complement** of 28°.

Answer: _____ °

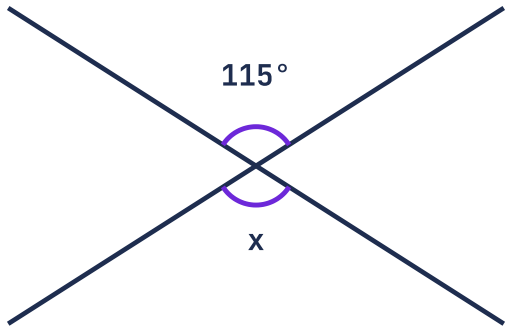
2 Score 3 NC.8.G.5

Find the **supplement** of 73°.

Answer: _____ °

3 Score 3 NC.8.G.5

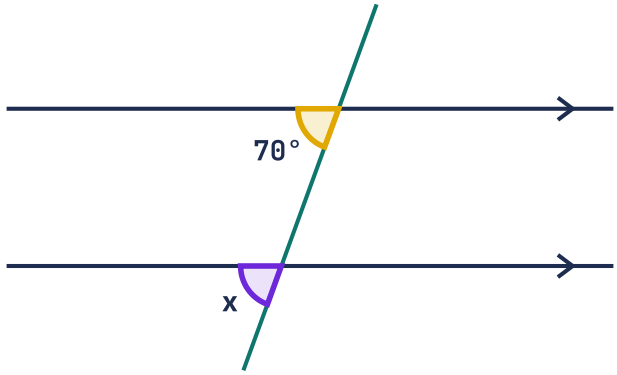
Two lines cross. Find **x**.



Answer: _____ °

4 Score 3 NC.8.G.5

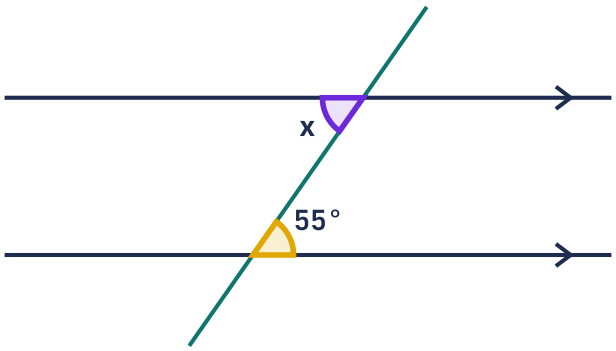
The lines are parallel. Find **x**.



Answer: _____ °

5 Score 3 NC.8.G.5

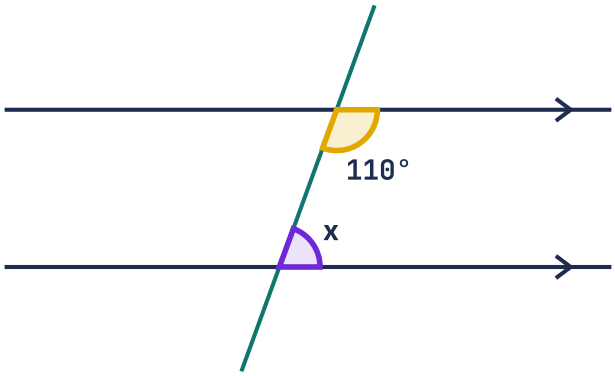
The lines are parallel. Find **x**.



Answer: _____ °

6 Score 3 NC.8.G.5

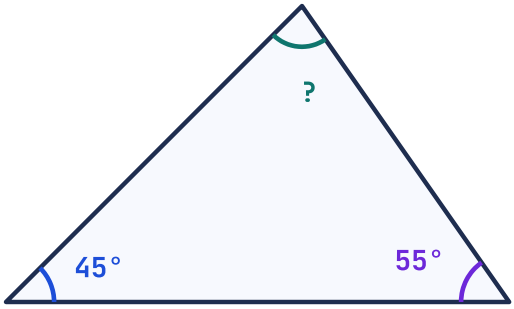
The lines are parallel. Find **x**.



Answer: _____ °

7 Score 3 NC.8.G.5

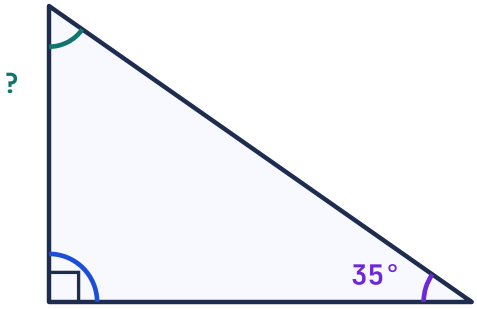
A triangle has angles 45° and 55°. Find the third angle.



Answer: _____ °

8 Score 3 NC.8.G.5

A right triangle has one angle of 35°. Find the other small angle.



Answer: _____ °

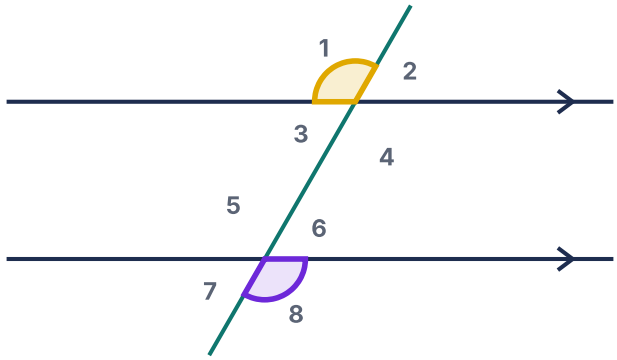
9 Score 3 NC.8.G.5

The far inside angles of a triangle are 40° and 65°. Find the exterior angle.

Answer: _____ °

10 Score 3 NC.8.G.5

What kind of pair are ∠1 and ∠8?



(A) Alternate interior angles

(B) Alternate exterior angles

(C) Corresponding angles

(D) Same-side interior angles

11 Score 3 NC.8.G.5

Find the mistake. Ava wrote: "The supplement of 40° is 50°." What went wrong?

(A) Ava used 90. Supplementary angles add to 180, so the answer is 140°.

(B) Ava is right: 40 + 50 = 90.

(C) It should be 40° because the angles are equal.

(D) It should be 320° because 360 − 40 = 320.

Pythagorean Theorem: Mini-Lessons

NC.8.G.6–8

Name _____ Date _____ Period _____

5.1 Legs, hypotenuse and $a^2 + b^2 = c^2$ NC.8.G.6

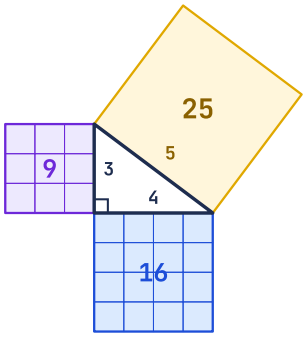
In a **right triangle**, the two shorter sides are the **legs** (a and b). The longest side, across from the right angle, is the **hypotenuse** (c). The theorem: **$a^2 + b^2 = c^2$**.

Right triangle
a triangle with a 90° angle (a square corner)

Leg
one of the two sides that make the square corner

Hypotenuse
the longest side, across from the square corner

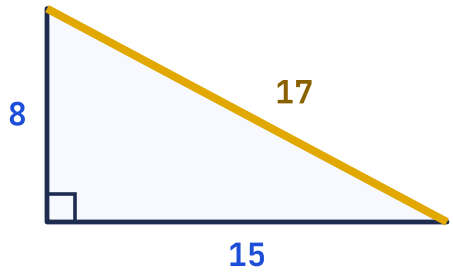
Watch out: The hypotenuse is **always** across from the square corner. Find it first.



$9 + 16 = 25$

$a^2 + b^2 = c^2$

Walk through: Which side is the hypotenuse?



1. Where is the right angle (square corner)?

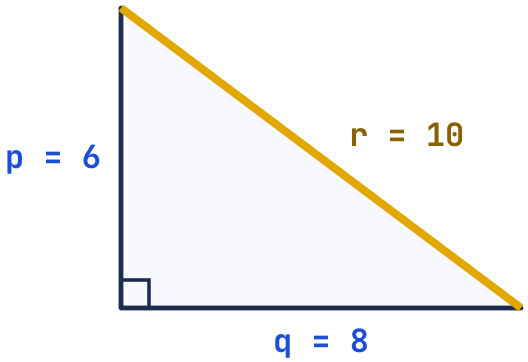
2. The hypotenuse is across from the right angle. Which side?

3. Check with the theorem: $8^2 + 15^2 = ?$

Check your understanding

1 Score 3 NC.8.G.6

Which side is the **hypotenuse**?



(A) r = 10
(B) p = 6
(C) q = 8

2 Score 3 NC.8.G.6

A right triangle has legs a and b and hypotenuse c. Which equation is true?

(A) $a^2 \cdot b^2 = c^2$
(B) $a^2 + c^2 = b^2$
(C) $a^2 + b^2 = c^2$
(D) $a + b = c$

5.2 Find the hypotenuse NC.8.G.7

When the **hypotenuse is missing**: square each leg, **add**, then take the **square root**.

Square root (√)
undoes squaring: √100 = 10

Round to the nearest tenth
one digit after the decimal point: √13 ≈ 3.6

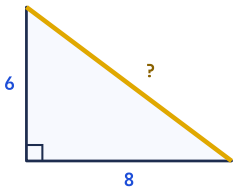
$6^2 + 8^2 = c^2$

$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

Watch out: Do not add the legs (6 + 8 = 14). Square them first.



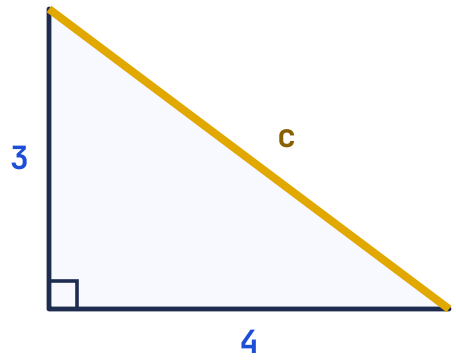
$6^2 + 8^2 = c^2$

$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

Walk through: `Legs 3 and 4. Find the hypotenuse.`



1. Write the theorem with the legs in place.

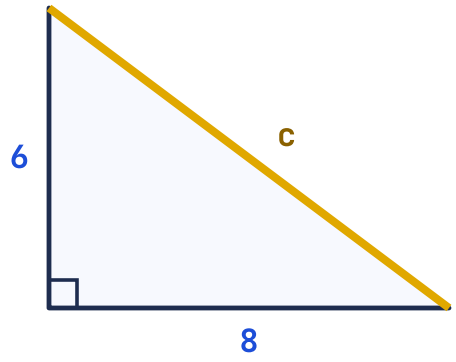
2. Square and add.

3. $c^2 = 25$. Find c.

Check your understanding

1 Score 3 NC.8.G.7

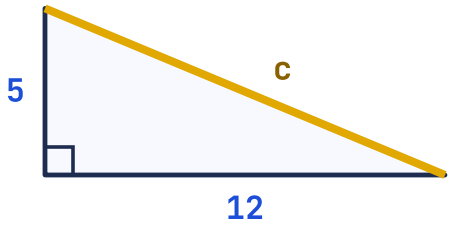
The legs of a right triangle are **6** and **8**. Find the hypotenuse.



Answer: _____

2 Score 3 NC.8.G.7

The legs of a right triangle are **5** and **12**. Find the hypotenuse.



Answer: _____

5.3 Find a missing leg NC.8.G.7

When a **leg is missing**: square the hypotenuse and the leg you know, **subtract**, then take the **square root**. `leg² = c² - (other leg)²`

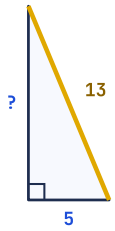
Hypotenuse → add

missing the long side: add the squares

Leg → subtract

missing a short side: subtract the squares

Watch out: If the missing side is a leg, the answer must be **shorter** than the hypotenuse. If it is not, you added by mistake.



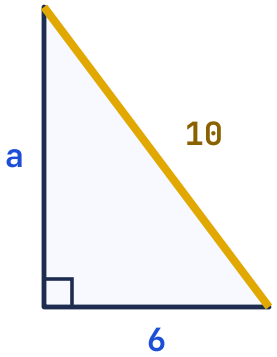
$a^2 + 5^2 = 13^2$

$a^2 + 25 = 169$

$a^2 = 169 - 25 = 144$

$a = \sqrt{144} = 12$

Walk through: **Hypotenuse 10, one leg 6. Find the other leg.**



1. Which side is missing?

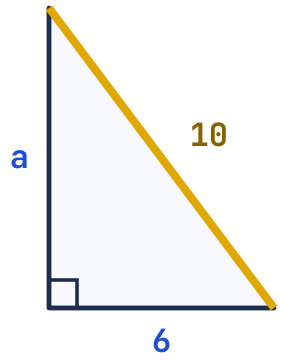
2. Square and subtract.

3. Find a.

Check your understanding

1 Score 3 NC.8.G.7

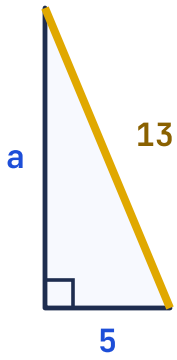
A right triangle has hypotenuse **10** and one leg **6**. Find the other leg.



Answer: _____

2 Score 3 NC.8.G.7

A right triangle has hypotenuse **13** and one leg **5**. Find the other leg.



Answer: _____

5.4 Is it a right triangle? (the converse) NC.8.G.6

To test three side lengths: square the two **shorter** sides and add. If the sum **equals** the longest side squared, it **is** a right triangle. If not, it is not.

Converse
the theorem used backward, to test a triangle

Longest side
the only side that could be the hypotenuse

Watch out: Always use the **longest** side as c.

`Sides 6, 8, 10` `36 + 64 = 100` `100 = 100 ✓` right triangle

`Sides 4, 5, 6` `16 + 25 = 41` `41 ≠ 36 ✗` not a right triangle

Walk through: `Do 9, 12, 15 make a right triangle?`

1. Which side would be the hypotenuse?

_____

2. Square and add the two shorter sides.

_____

3. $15^2 = 225$. What do we decide?

_____

Check your understanding

1 Score 3 NC.8.G.6

Can the sides **6, 8, 10** make a right triangle?

(A) Yes: 36 + 64 = 100, which is 10^2

(B) No: the numbers are not 3, 4, 5

(C) No: 6 + 8 is not 10

2 Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

(A) No: 16 + 25 = 41, but $6^2 = 36$

(B) Yes: 4 + 5 is more than 6

(C) Yes: they are in counting order

5.5 Distance on the coordinate plane NC.8.G.8

To find the distance between two points: draw a right triangle. The **across** change is one leg, the **up/down** change is the other leg, and the **distance** is the hypotenuse.

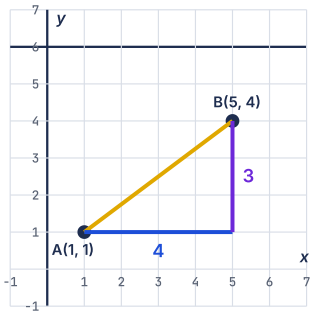
Run

the across change: subtract the x-values

Rise

the up/down change: subtract the y-values

Watch out: Count the length of each leg (always positive). Then use $a^2 + b^2 = c^2$.




```
run = 5 - 1 = 4
rise = 4 - 1 = 3
4² + 3² = 16 + 9 = 25
distance = √25 = 5
```

Walk through: `Find the distance from (2, 1) to (8, 9).`

1 Find the run (across change).

2 Find the rise (up/down change).

3 Use $a^2 + b^2 = c^2$.

Check your understanding

1 Score 3 NC.8.G.8

Find the distance from **(1, 2)** to **(4, 6)**.

Answer: _____

SKILL BUILDERS · UNIT 5 · SCORE 3

Pythagorean Theorem: Practice Set

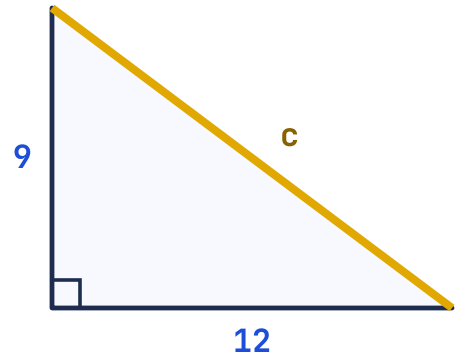
NC.8.G.6–8

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.7

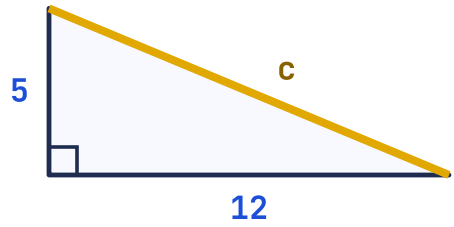
The legs of a right triangle are **9** and **12**. Find the hypotenuse.



Answer: _____

2 Score 3 NC.8.G.7

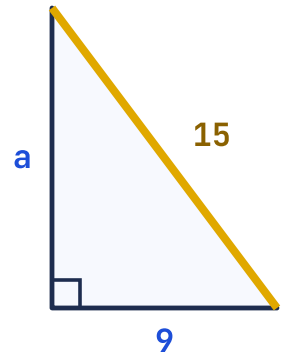
The legs of a right triangle are **5** and **12**. Find the hypotenuse.



Answer: _____

3 Score 3 NC.8.G.7

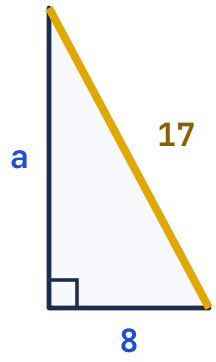
A right triangle has hypotenuse **15** and one leg **9**. Find the other leg.



Answer: _____

4 Score 3 NC.8.G.7

A right triangle has hypotenuse **17** and one leg **8**. Find the other leg.



Answer: _____

5 Score 3 NC.8.G.6

Can the sides **7, 24, 25** make a right triangle?

(A) Yes: 49 + 576 = 625, which is 25^2

(B) No: the numbers are not 3, 4, 5

(C) No: 7 + 24 is not 25

6 Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

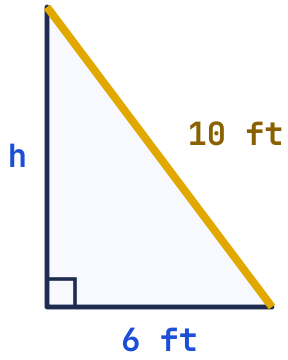
(A) No: 16 + 25 = 41, but $6^2 = 36$

(B) Yes: 4 + 5 is more than 6

(C) Yes: they are in counting order

7 Score 3 NC.8.G.7

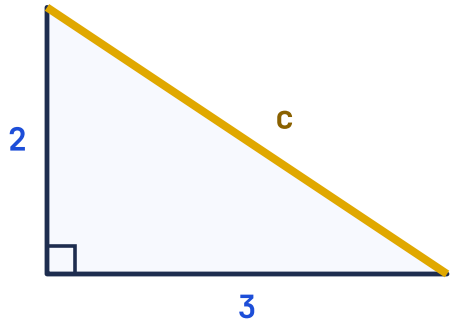
A 10-foot ladder leans on a wall. The bottom is **6 feet** from the wall. How high up the wall does it reach?



Answer: _____ ft

8 Score 3 NC.8.G.7

The legs of a right triangle are **2** and **3**. Find the hypotenuse to the nearest tenth.



Answer: _____

9 Score 3 NC.8.G.8

Find the distance from **(−2, 1)** to **(2, 4)**.

Answer: _____

10 Score 3 NC.8.G.7

Find the mistake. A right triangle has legs 6 and 8. Max wrote: `6 + 8 = 14, so c = 14`. What went wrong?

(A) Max is right: c = 14.

(B) Max added the legs. He must square them: 36 + 64 = 100, so c = 10.

(C) Max should subtract: 8 − 6 = 2.

(D) Max should multiply: 6 × 8 = 48.

Functions: Mini-Lessons

NC.8.F.1–5 · NC.8.EE.8

Name _____ Date _____ Period _____

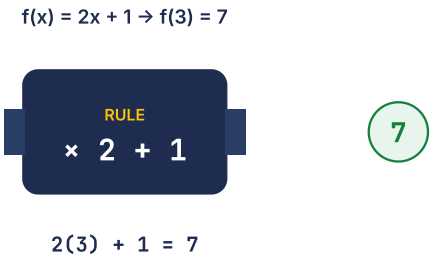
6.1 What is a function? NC.8.F.1

A **function** is a rule where **each input (x) has exactly one output (y)**. f(x) means "the output when the input is x." To find f(4), put 4 in for x.

Input (x)
the number you put in

Output (y or f(x))
the number that comes out

Function
every input has exactly one output



Not a function: one input with **two** outputs, like x = 2 going to both 5 and 7. On a graph, a vertical line that hits the graph twice means it is **not** a function.

Watch out: Two inputs with the same output is still a function. The problem is one input with two outputs.

Walk through: `f(x) = 3x - 2. Find f(4).`

1. What does f(4) ask?

2. Multiply first: 3(4) = ?

3. Finish: 12 − 2 = ?

Check your understanding

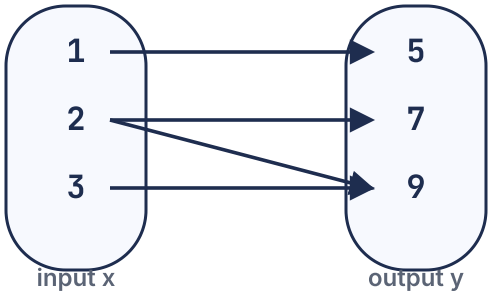
1 Score 3 NC.8.F.1

`f(x) = 2x + 5`. Find `f(3)`.

Answer: _____

2 Score 3 NC.8.F.1

Is this mapping a function?



Ⓐ No: the input 2 has two outputs

Ⓑ Yes: every input has an arrow

Ⓒ Yes: two inputs share an output

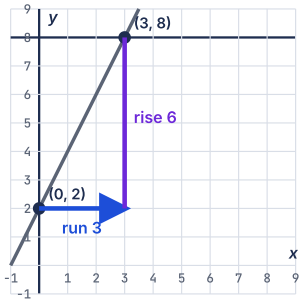
6.2 Slope: rise over run NC.8.F.4

Slope tells how steep a line is. `slope = rise ÷ run = (change in y) ÷ (change in x)`. Uphill (left to right) is **positive**, downhill is **negative**, flat is **0**.

Rise
the up/down change: subtract the y-values

Run
the across change: subtract the x-values

Slope (m)
rise ÷ run



`slope = rise ÷ run`

`= (8 - 2) ÷ (3 - 0)`

`= 6 ÷ 3 = 2`

up 2 for every 1 across

Green: positive (uphill) · Red: negative (downhill) · Gray: zero (flat)

Watch out: Rise goes on **top**. Run goes on the bottom. And subtract in the same order for both.

Walk through: `Find the slope from (1, 3) to (4, 12).`

1 Find the rise (change in y).

_____

2 Find the run (change in x).

3 Slope = rise ÷ run = ?

Check your understanding

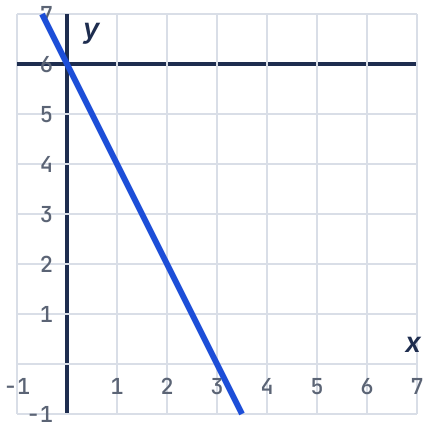
1 Score 3 NC.8.F.4

Find the slope of the line through **(0, 2)** and **(3, 8)**.

Answer: _____

2 Score 3 NC.8.F.4

What kind of slope does this line have?



Ⓐ Negative

Ⓑ Positive

Ⓒ Zero

6.3 y = mx + b from a table or graph NC.8.F.4

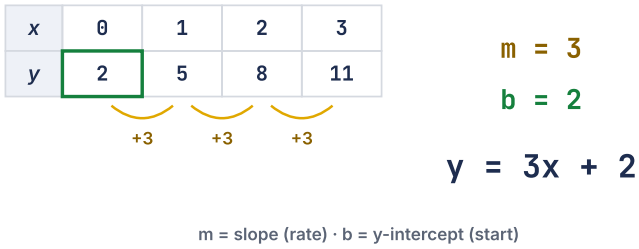
A line can be written **y = mx + b**. **m** is the slope (how much y changes when x goes up 1). **b** is the y-intercept (the y-value when x = 0, where the line crosses the y-axis).

Slope-intercept form

y = mx + b

y-intercept (b)

where the line crosses the y-axis; the y-value when x = 0



x	0	1	2	3
y	5	8	11	14

y goes up 3 each time → m = 3. When x = 0, y = 5 → b = 5. So y = 3x + 5.

Watch out: In y = 4x − 1, the slope is 4 and the y-intercept is **−1** (keep the sign).

Walk through: Write the equation for this table.

1

x	0	1	2	3
y	1	3	5	7

How much does y change when x goes up 1?

2 What is y when x = 0?

3 Write y = mx + b.

Check your understanding

1 Score 3 NC.8.F.3

In **y = 4x - 1**, what is the **slope**?

Answer: _____

2 Score 3 NC.8.F.4

Which equation matches the table?

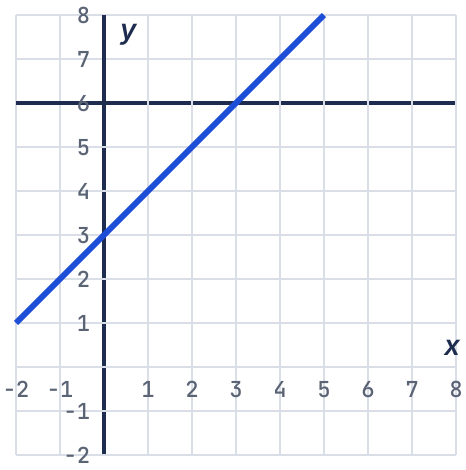
x	0	1	2	3
y	4	6	8	10

(A) y = 2x + 4 (B) y = 2x

(C) y = 4x + 2 (D) y = x + 4

3 Score 3 NC.8.F.4

What is the y-intercept of this line?



Answer: _____

6.4 Rate and start value NC.8.F.2

In a real-world line, the **slope is the rate** (per hour, per mile, each day) and the **y-intercept is the start value** (the amount when x = 0, like a fee).

Rate of change
how much y changes for each 1 of x

Initial (start) value
the amount at the start, when x = 0

Increasing / decreasing
a graph going up / going down from left to right



`Taxi: y = 2x + 3`

2 = rate: $2 per mile

3 = start: $3 before driving

`5 miles: 2(5) + 3 = $13`

Phone plan: **`y = 15x + 40`**. $40 to start, then $15 each month.

Watch out: The start value is the number **by itself**, not the number next to x.

Walk through: **`A gym charges y = 20x + 50 for x months. What does 50 mean?`**

1 In y = 20x + 50, which number is alone (not next to x)?

_____

2 What does the number alone mean?

_____

3 And 20?

Check your understanding

1 Score 3 NC.8.F.2

A plant is $h = 2w + 6$ cm tall after w weeks. How tall was it at the start?

Answer: _____ cm

2 Score 3 NC.8.F.2

A phone plan costs $y = 15x + 40$. What is the cost **per month**?

Answer: _____ dollars

3 Score 3 NC.8.F.5

A graph of water in a tank goes **down** from left to right. What is happening?

Ⓐ The water is decreasing

Ⓑ The water is increasing

Ⓒ The water stays the same

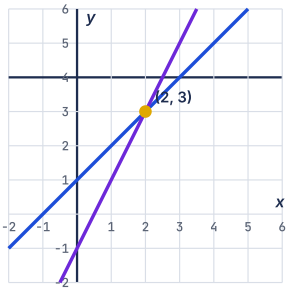
6.5 Systems: where two lines cross NC.8.EE.8

The **solution of a system** is the point **(x, y) where the two lines cross**. It makes **both** equations true.

System of equations
two equations used together

Solution
the point that is on both lines

Watch out: A point on only one line is not the solution. Check it in **both** equations.



$y = x + 1$

$y = 2x - 1$

Check (2, 3):

$3 = 2 + 1$ ✓

$3 = 2(2) - 1$ ✓

Walk through: Is (1, 3) the solution of y = x + 2 and y = 3x?

1 Check the first equation: y = x + 2 with x = 1.

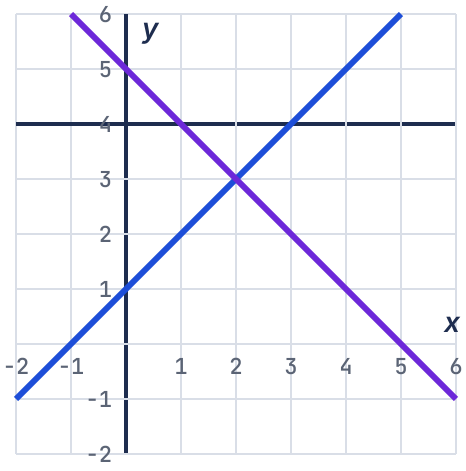
2 Check the second equation: y = 3x with x = 1.

3 Is (1, 3) the solution?

Check your understanding

1 Score 3 NC.8.EE.8

What is the solution of the system shown?



(A) (0, 1)

(B) (2, 3)

(C) (3, 2)

(D) (0, 5)

2 Score 3 NC.8.EE.8

Is **(2, 5)** the solution of **y = x + 3** and **y = 2x**?

(A) No: it works in y = x + 3, but 2(2) = 4, not 5

(B) Yes: 2 + 3 = 5

(C) Yes: it works in both

SKILL BUILDERS · UNIT 6 · SCORE 3

Functions: Practice Set

NC.8.F.1–5 · NC.8.EE.8

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.F.1

$f(x) = 4x + 1$. Find $f(2)$.

Answer: _____

2 Score 3 NC.8.F.1

Which table shows a function?

(A)

x	1	1	2	3
y	4	5	6	7

(B)

x	2	3	2	4
y	1	2	3	4

(C)

x	1	2	3	4
y	5	5	6	6

3 Score 3 NC.8.F.4

Find the slope from **(1, 3)** to **(4, 12)**.

Answer: _____

4 Score 3 NC.8.F.4

Find the slope from **(0, 6)** to **(3, 0)**.

Answer: _____

5 Score 3 NC.8.F.4

Which equation matches the table?

x	0	1	2
y	5	8	11

(A) y = 3x

(B) y = 3x + 5

(C) y = 5x + 3

(D) y = x + 5

6 Score 3 NC.8.F.2

A savings account has **y = 5x + 20** dollars after x weeks. What is the **start value**?

Answer: _____ dollars

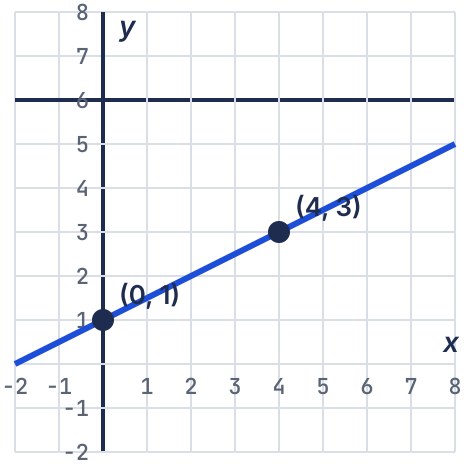
7 Score 3 NC.8.F.2

In **y = 5x + 20**, what is the **rate** (amount added each week)?

Answer: _____ dollars

8 Score 3 NC.8.F.4

What is the slope of this line?



Answer: _____

9 Score 3 NC.8.EE.8

Solve the system: **y = x + 2** and **y = 3x**.

(A) (1, 3)

(B) (2, 4)

(C) (3, 1)

(D) (0, 2)

10 Score 3 NC.8.F.5

A runner's distance-from-home graph is **flat** for a while. What is the runner doing?

(A) Stopped: the distance is not changing

(B) Running faster

(C) Running home

11 Score 3 NC.8.F.4

Find the mistake. For (2, 1) and (4, 7), Kai wrote: `slope = (4 - 2) ÷ (7 - 1) = 2 ÷ 6 = 1/3`. What went wrong?

(A) Kai put the run on top. Slope = rise ÷ run = 6 ÷ 2 = 3.

(B) Kai is right.

(C) The slope should be negative.

(D) Kai should add: (4 + 2) ÷ (7 + 1).

Statistics: Mini-Lessons

NC.8.SP.1–4

Name _____ Date _____ Period _____

7.1 Scatter plots and correlation NC.8.SP.1

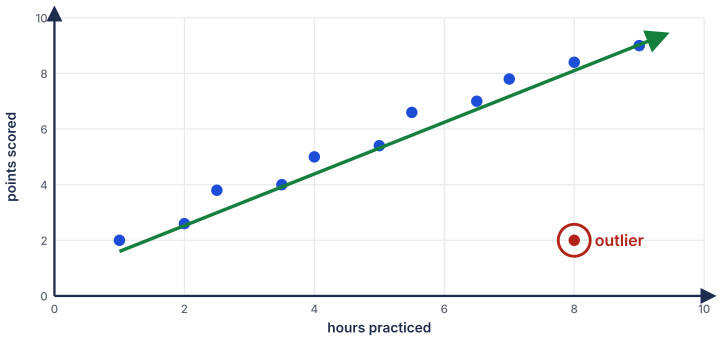
A **scatter plot** shows two sets of data as dots. Describe it: **direction** (positive = up, negative = down, none), **strength** (close to a line = strong), and **outliers** (dots far from the rest).

Positive correlation
as x goes up, y goes up

Negative correlation
as x goes up, y goes down

Outlier
a dot far away from the pattern

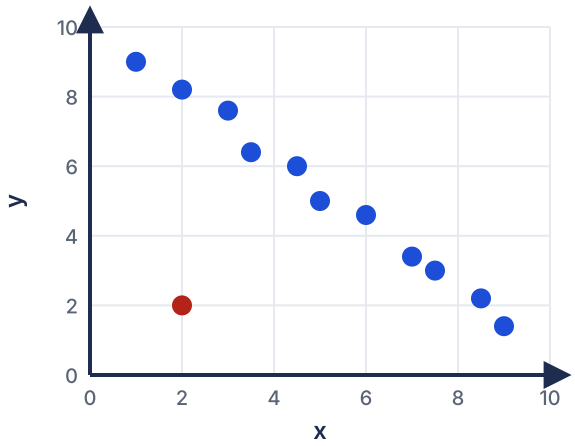
Correlation is not causation
two things moving together does not prove one causes the other



Left: negative (down). Right: no correlation (no pattern).

Watch out: Ice cream sales and sunburns both go up in summer. Ice cream does not **cause** sunburns: hot weather drives both.

Walk through: `Describe this scatter plot.`



1 Read left to right. Which way do most dots go?

_____

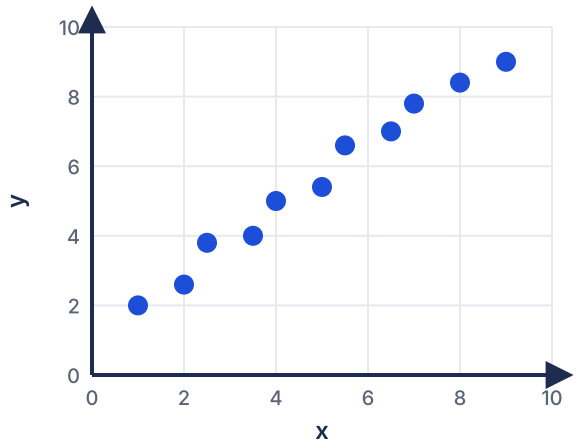
2 Are the dots close to a line or spread out?

3 Is there an outlier?

Check your understanding

1 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



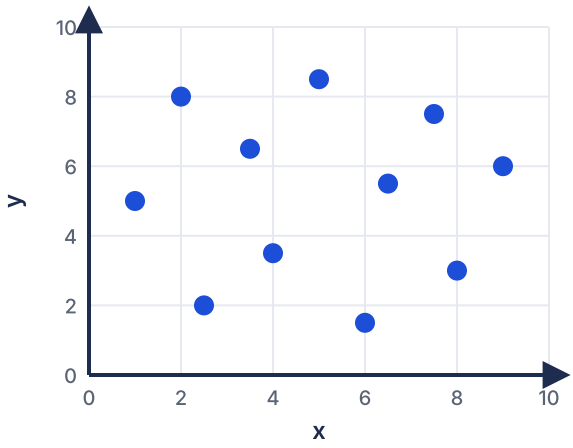
(A) Positive correlation

(B) Negative correlation

(C) No correlation

2 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



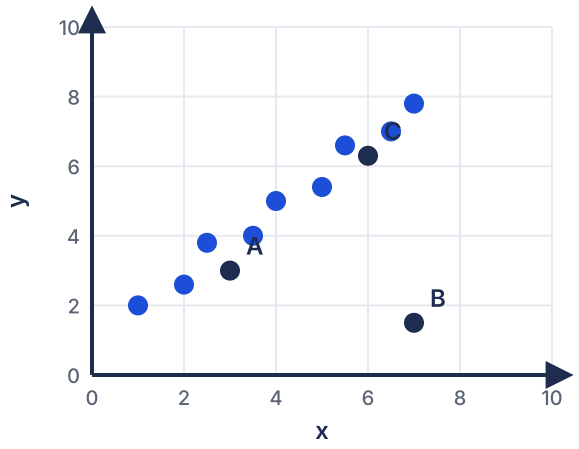
(A) No correlation

(B) Positive correlation

(C) Negative correlation

3 Score 3 NC.8.SP.1

Which point is an **outlier**?



(A) B

(B) A

(C) C

7.2 Line of best fit and predictions NC.8.SP.2

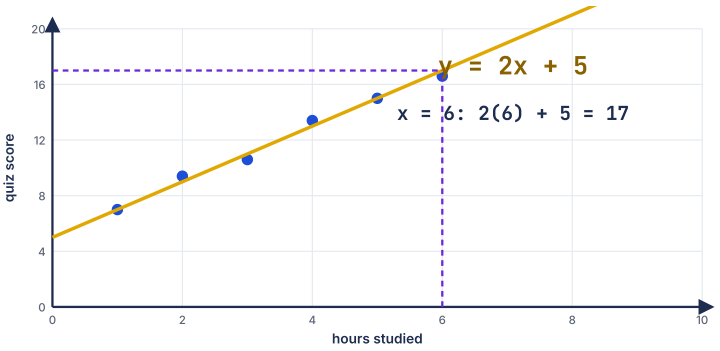
A **line of best fit** is a straight line through the middle of the dots. It is just **y = mx + b**: plug in an x to **predict** y.

Line of best fit
a line close to most dots, about as many above as below

Interpolation
predicting inside the data (safer)

Extrapolation
predicting outside the data (riskier)

Watch out: A good fit line does not have to touch any dot. It goes through the middle of the group.



Walk through:

Best fit: y = 3x + 4 (x = hours, y = score). Predict y for x = 5.

1. What do we do with x = 5?

2. 3(5) = ?

3. Finish.

Check your understanding

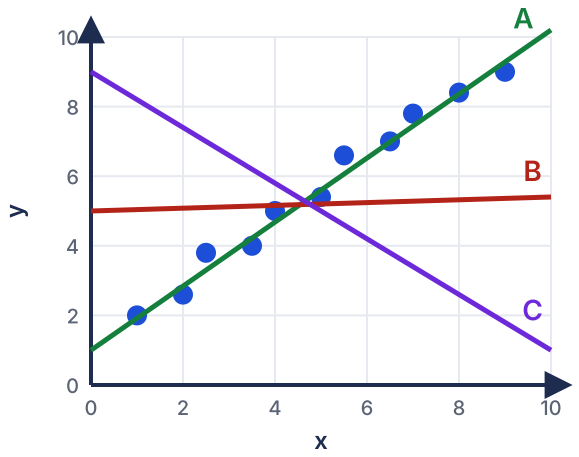
1 Score 3 NC.8.SP.3

Best fit: **y = 2x + 5**. Predict y when x = 6.

Answer: _____

2 Score 3 NC.8.SP.2

Which line fits the data best?



Ⓐ Line A

Ⓑ Line B

Ⓒ Line C

7.3 Two-way tables NC.8.SP.4

A **two-way table** counts data sorted two ways. Inside boxes are **joint** counts; the edge totals are **marginal** counts. **Relative frequency** = count ÷ total, often as a percent.

Joint frequency

one inside box

Marginal frequency

a row or column total

Relative frequency

a count divided by a total: 12 ÷ 40 = 30%

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

	Likes dogs	Does not	Total
Boys	12	8	20
Girls	15	5	20
Total	27	13	40

12 ÷ 20 = 0.60 = 60%

Watch out: Always ask "out of which total?" Out of everyone (40), or out of one row (20)?

Walk through:

Use the table. What percent of all students are girls who like dogs?

1

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

Find the box: girls who like dogs.

_____

2 Out of which total?

_____

3 15 ÷ 40 = ?

_____

Check your understanding

1 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

How many **boys** are there in all?

Answer: _____

2 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

What percent of **all** students do **not** like dogs?

Answer: _____ %

SKILL BUILDERS · UNIT 7 · SCORE 3

Statistics: Practice Set

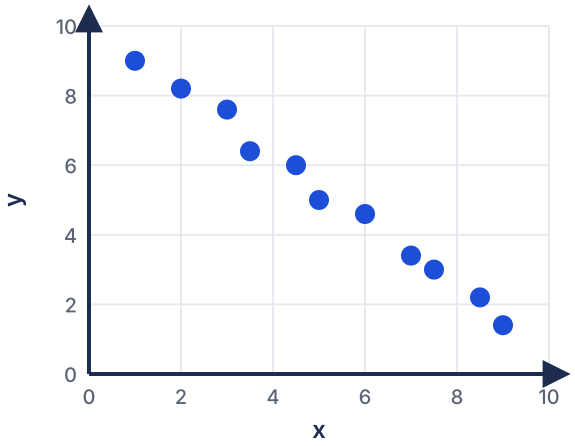
NC.8.SP.1–4

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



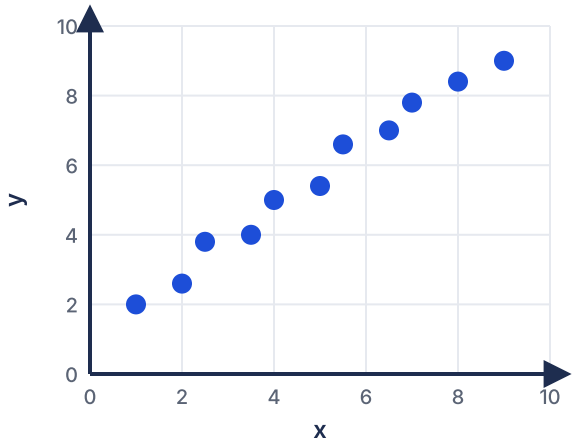
(A) Negative correlation

(B) Positive correlation

(C) No correlation

2 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



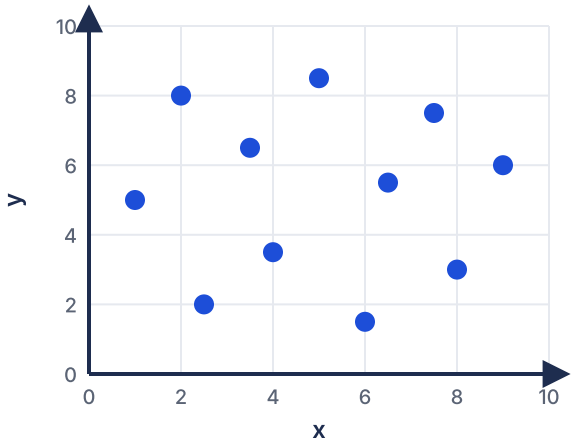
(A) Positive correlation

(B) Negative correlation

(C) No correlation

3 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



(A) No correlation

(B) Positive correlation

(C) Negative correlation

4 Score 3 NC.8.SP.1

A single dot far away from the rest of a scatter plot is called...

(A) an outlier

(B) a line of best fit

(C) a correlation

5 Score 3 NC.8.SP.3

Best fit: $y = 3x + 4$. Predict y when x = 5.

Answer: _____

6 Score 3 NC.8.SP.3

A best fit line for plant height is $y = 2x + 3$ (x = weeks, y = cm). What does the slope 2 mean?

(A) The plant is 2 weeks old

(B) The plant grows about 2 cm each week

(C) The plant started at 2 cm

7 Score 3 NC.8.SP.2

Data go from x = 1 to x = 8. Using the line to predict y at x = 30 is...

Ⓐ extrapolation (riskier)

Ⓑ interpolation (safer)

8 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

How many **girls** like dogs?

Answer: _____

9 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

Of the **boys**, what percent like dogs?

Answer: _____ %

10 Score 3 NC.8.SP.1

Find the mistake. A scatter plot shows that towns with more ice cream shops have more sunburns. Rio says: "Ice cream causes sunburns." What is wrong?

Ⓐ Nothing. The data prove it.

Ⓑ A correlation does not prove cause. Hot, sunny weather can cause both.

Ⓒ The correlation must be negative.

Ⓓ Sunburns cause ice cream shops.

Transformations: Mini-Lessons

NC.8.G.2–4

Name _____ Date _____ Period _____

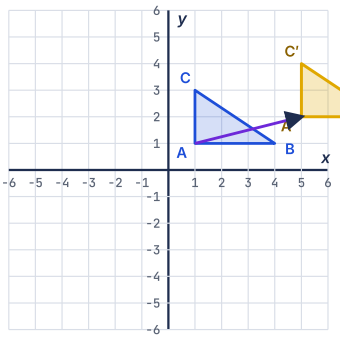
8.1 Translations (slides) NC.8.G.3

A **translation** slides a figure. **Right/left changes x** (add or subtract). **Up/down changes y**. The image is **congruent** (same size and shape).

Pre-image
the original figure

Image
the figure after the move, named with ′ (A′, "A prime")

Congruent
same size and same shape



A(1, 1) → A′(5, 2)
B(4, 1) → B′(8, 2)
C(1, 3) → C′(5, 4)

(x, y) → (x + 4, y + 1)

same size and shape: congruent

Move	right h	left h	up k	down k
Do	x + h	x − h	y + k	y − k

Watch out: Left and right change **x** (the first number). Up and down change **y** (the second number).

Walk through: `Translate (1, -2) left 3 and up 5.`

1. Left 3 changes which number?

_____

2. Up 5 changes which number?

_____

3. The image point is...

_____

Check your understanding

1 Score 3 NC.8.G.3

Translate right 4 and up 1: the point **(2, 3)**. Where does it land?

Ⓐ (−2, 4) Ⓑ (3, 7)

Ⓒ (6, 4) Ⓓ (6, 2)

2 Score 3 NC.8.G.3

Translate left 2 and down 3: the point **(2, 5)**. Where does it land?

Ⓐ (0, 8) Ⓑ (−1, 3)

Ⓒ (4, 8) Ⓓ (0, 2)

8.2 Reflections (flips) NC.8.G.3

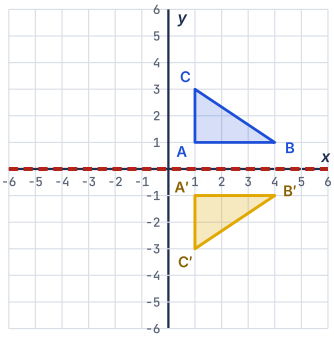
A **reflection** flips a figure over a line. Over the **x-axis**: y changes sign, `(x, y) → (x, -y)`. Over the **y-axis**: x changes sign, `(x, y) → (-x, y)`.

Line of reflection

the mirror line

Opposite

same number, other sign: the opposite of 5 is −5



`A(1, 1) → A'(1, -1)`
`B(4, 1) → B'(4, -1)`
`C(1, 3) → C'(1, -3)`

`(x, y) → (x, -y)`

over the x-axis: y changes sign

Trick: you flip **over** the x-axis, so the **other** letter (y) changes sign.

Watch out: Only **one** coordinate changes sign in a reflection over an axis.

Walk through: `Reflect (3, 5) over the y-axis.`

1 Over the y-axis, which coordinate changes sign?

2 The image is...

Check your understanding

1 Score 3 NC.8.G.3

Reflect over the x-axis: the point **(3, 5)**. Where does it land?

Ⓐ (3, −5) Ⓑ (−3, 5)

Ⓒ (5, 3) Ⓓ (−3, −5)

2 Score 3 NC.8.G.3

Reflect over the y-axis: the point **(−3, 4)**. Where does it land?

Ⓐ (3, 4) Ⓑ (4, −3)

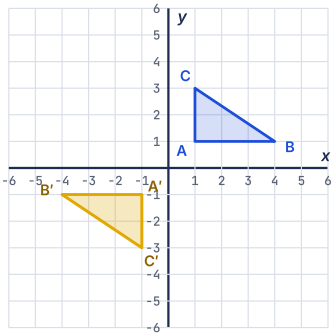
Ⓒ (3, −4) Ⓓ (−3, −4)

8.3 Rotations (turns) NC.8.G.3

A **rotation** turns a figure around the origin. **180°**: `(x, y) → (-x, -y)`. **90° counterclockwise**: `(x, y) → (-y, x)`. **270° counterclockwise** (90° clockwise): `(x, y) → (y, -x)`.

Counterclockwise
the opposite way a clock's hands turn

Origin
the point (0, 0) where the axes cross



`A(1, 1) → A'(-1, -1)`
`B(4, 1) → B'(-4, -1)`
`C(1, 3) → C'(-1, -3)`

`(x, y) → (-x, -y)`

180° turn: both signs change

Turn	Rule	(4, 2) →
90° CCW	(−y, x)	(−2, 4)
180°	(−x, −y)	(−4, −2)
270° CCW	(y, −x)	(2, −4)

Watch out: For 90° and 270°, the numbers **switch places**. For 180°, they do not.

Walk through: `Rotate (1, 5) 90° counterclockwise.`

1 Which rule is 90° counterclockwise?

2 Here x = 1 and y = 5. What is −y?

3 The image is...

Check your understanding

1 Score 3 NC.8.G.3

Rotate 180° about the origin: the point **(4, 2)**. Where does it land?

Ⓐ (2, 4) Ⓑ (−4, −2)
Ⓒ (−4, 2) Ⓓ (−2, 4)

2 Score 3 NC.8.G.3

Rotate 90° counterclockwise about the origin: the point **(5, 5)**. Where does it land?

Ⓐ (5, 5) Ⓑ (−5, −5)
Ⓒ (5, −5) Ⓓ (−5, 5)

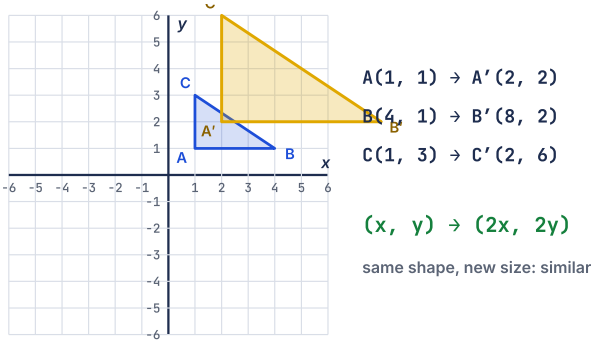
8.4 Dilations, congruent and similar NC.8.G.4

A **dilation** multiplies both coordinates by the **scale factor k**: `(x, y) → (kx, ky)`. k > 1 makes it bigger; k between 0 and 1 makes it smaller. The image is **similar**. Slides, flips and turns keep the image **congruent**.

Scale factor (k)
the number you multiply by

Similar
same shape, maybe a different size

Congruent
same shape AND same size



Dilation = resize = similar. Translation, reflection, rotation = same size = congruent.

Watch out: Multiply **both** numbers by k. Do not add k.

Walk through: `Dilate (3, -4) by k = 3.`

1. What do we do with k = 3?

2. Multiply.

3. Is the image congruent or similar?

Check your understanding

1 Score 3 NC.8.G.4

Dilate by k = 2: the point **(2, 3)**. Where does it land?

(A) (4, 6)
(B) (4, 5)
(C) (1, 1.5)
(D) (2, 6)

2 Score 3 NC.8.G.4

After a **dilation** with k = 3, the image is...

(A) similar to the pre-image
(B) congruent to the pre-image
(C) neither congruent nor similar

3 Score 3 NC.8.G.2

After a **reflection**, the image is...

Ⓐ congruent to the pre-image

Ⓑ bigger than the pre-image

Ⓒ smaller than the pre-image

SKILL BUILDERS · UNIT 8 · SCORE 3

Transformations: Practice Set

NC.8.G.2–4

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.3

Translate left 2 and down 3: the point **(2, 5)**. Where does it land?

(A) (−1, 3)
(B) (0, 2)
(C) (4, 8)
(D) (0, 8)

2 Score 3 NC.8.G.3

Reflect over the y-axis: the point **(−3, 4)**. Where does it land?

(A) (4, −3)
(B) (−3, −4)
(C) (3, −4)
(D) (3, 4)

3 Score 3 NC.8.G.3

Reflect over the x-axis: the point **(6, −1)**. Where does it land?

(A) (6, 1)
(B) (−6, 1)
(C) (−1, 6)
(D) (−6, −1)

4 Score 3 NC.8.G.3

Rotate 180° about the origin: the point **(2, 7)**. Where does it land?

(A) (−2, −7)
(B) (−2, 7)
(C) (−7, 2)
(D) (7, 2)

5 Score 3 NC.8.G.4

Dilate by k = 4: the point **(1, 3)**. Where does it land?

(A) (4, 3)
(B) (4, 12)
(C) (5, 7)
(D) (0.25, 0.75)

6 Score 3 NC.8.G.3

Rotate 90° counterclockwise about the origin: the point **(5, 5)**. Where does it land?

(A) (−5, −5)
(B) (5, −5)
(C) (5, 5)
(D) (−5, 5)

7 Score 3 NC.8.G.4

Dilate by k = 1/2: the point **(4, −6)**. Where does it land?

Ⓐ (4.5, −5.5)

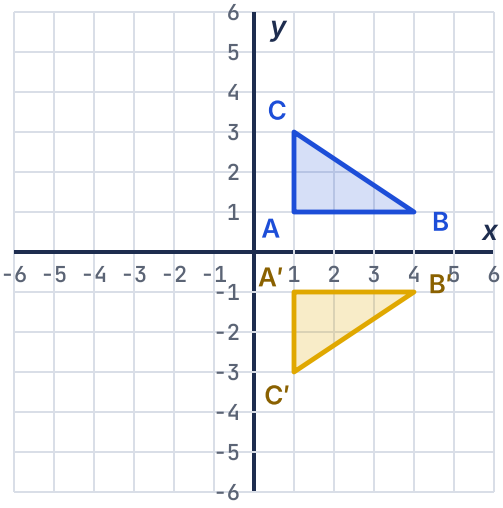
Ⓑ (2, −3)

Ⓒ (2, 3)

Ⓓ (8, −12)

8 Score 3 NC.8.G.3

Which move takes the blue triangle to the gold triangle?



Ⓐ Translation down 4

Ⓑ Reflection over the x-axis

Ⓒ Dilation by k = 2

Ⓓ Rotation 180°

9 Score 3 NC.8.G.2

After a translation, a reflection and a rotation, the image is...

Ⓐ congruent: same size and shape

Ⓑ similar but smaller

Ⓒ similar but bigger

10 Score 3 NC.8.G.3

First translate right 3, then reflect over the x-axis: the point **(1, 2)**. Where does it land?

Ⓐ (4, 2)

Ⓑ (4, −2)

Ⓒ (−4, −2)

Ⓓ (1, −5)

11 Score 3 NC.8.G.3

Find the mistake. Mia reflected (2, 6) over the x-axis and got **`(-2, 6)`**. What went wrong?

Ⓐ Mia changed x. Over the x-axis, y changes sign: (2, −6).

Ⓑ Mia is right.

Ⓒ The numbers should switch: (6, 2).

Ⓓ Both signs should change: (−2, −6).